

Stage 1.4.2.5.4 — Docling Table / Image / Formula Understanding

After Level 3 — Image Content Understanding & Semantic Extraction, I recommend that we finish Stage 1.4.2.5.4 by covering tables and formulas systematically, rather than moving immediately to 1.4.2.5.5.

Our learning path can now be:

```

Stage 1.4.2.5.4
Docling Table / Image / Formula Understanding
├── Level 1
│   ├── Picture Detection
│   └── [x] Completed
├── Level 2
│   ├── Picture Extraction
│   └── [x] Completed
├── Level 3
│   ├── Image Content Understanding & Semantic Extraction
│   └── [*] Current / completing validation
├── Level 4
│   ├── Table Understanding & Structured Extraction
│   └── [>] Next
├── Level 5
│   ├── Formula Understanding & Extraction
│   └── [>]
└── Level 6
    ├── Unified Complex-Content Representation
    └── [>]
  
```

What each remaining level teaches

Level 4 — Table Understanding & Structured Extraction

We will test whether Docling can preserve:

```

PDF Table
  ↓
Docling TableItem
  ↓
Rows / columns / cells
  ↓
Structured representation
  ↓
Markdown / HTML / DataFrame-style representation
  
```

We'll also deliberately test the issue we encountered earlier: table vs. visual layout, including a table positioned beside another content region.

Level 5 — Formula Understanding & Extraction

Our sample PDF already contains a mathematical formula.

We'll investigate:

```
Formula image / PDF formula
  ↓
Docling
  ↓
Formula detected?
  ↓
LaTeX / semantic representation?
```

We'll distinguish:

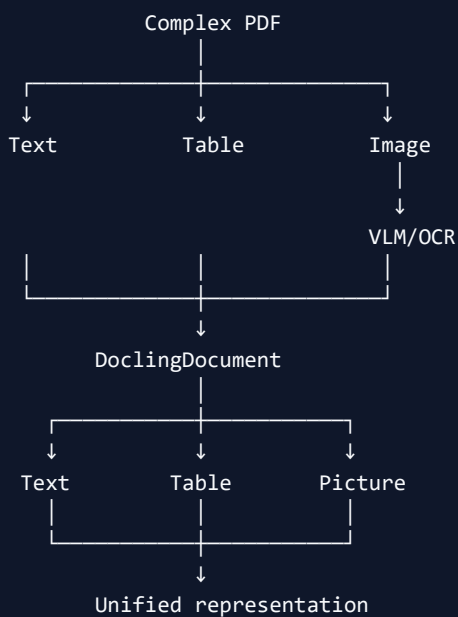
Formula as an image

≠

Formula recognized as a mathematical expression

Level 6 — Unified Complex-Content Representation

This is the important final exercise before moving to LangChain. We'll take our complete PDF:



Then we'll ask the most important RAG question:

Can we represent all these different content types in a form that a downstream RAG pipeline can actually consume?

Then Stage 1.4.2.5.4 is complete

At that point:

Stage 1.4.2.5.4
Docling Table / Image / Formula Understanding



Stage 1.4.2.5.5
Docling → LangChain → RAG

And 1.4.2.5.5 will be where everything starts coming together:

```
graph TD; A[PDF] --> B[Docling]; B --> C[DoclingDocument]; C --> D[LangChain Documents]; D --> E[Metadata]; E --> F[Chunking]; F --> G[Embeddings]; G --> H[Vector Store]; H --> I[Retrieval]; I --> J[RAG];
```

So, immediately next

We should finish Level 3 validation first, because our FOUND result has not yet proven that the words came from the architecture image.

Then:

Level 4 — Table Understanding & Structured Extraction

That gives us a clean progression:

Image → Understand → Table → Formula → Unified representation → LangChain → RAG.